

MATH – 241
Engineering Mathematics I
Problem Set – 3

Question 1. Give an example of non-empty subset $\mathbf{U}$ of $\mathbb{R}^2$ such that $\mathbf{U}$ is closed under addition and under taking additive inverses, i.e, $\forall \underline{u} \in \mathbf{U}, -\underline{u} \in \mathbf{U}$ but $\mathbf{U}$ is not a subspace of $\mathbb{R}^2$.

Question 2. Give an example of non-empty subset $\mathbf{U}$ of $\mathbb{R}^2$ such that $\mathbf{U}$ is closed under scalar multiplication, but $\mathbf{U}$ is not a subspace of $\mathbb{R}^2$.

Question 3. Suppose $\{\underline{v}_1, \dots, \underline{v}_n\}$ is a linearly independent set in $\mathbf{V}$ and $\underline{w} \in \mathbf{V}$. Prove that if $\{\underline{v}_1 + \underline{w}, \dots, \underline{v}_n + \underline{w}\}$ is linearly dependent then $\underline{w} \in \text{Span}\{\underline{v}_1, \dots, \underline{v}_n\}$.

Question 4. Let $\mathbf{U}$ be a subspace of $\mathbb{R}^5$ defined by:

$$\mathbf{U} = \left\{ \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} : u_1 = 3u_2 \text{ and } u_3 = 7u_4 \right\}$$

Find a basis for $\mathbf{U}$.

Question 5. Find a basis $\{p_0, p_1, p_2, p_3\}$ of

$\mathbf{P}_3 = \{\text{all real coefficient polynomials of degree less than or equal to 3}\}$,
such that none of the polynomials has degree 0 or 2.

Question 6. Prove that in a vector space $\mathbf{V}$, $\text{Span}\{\underline{v}_1, \dots, \underline{v}_k\}$ is the smallest subspace that contains $\{\underline{v}_1, \dots, \underline{v}_k\}$.